

Fedi Khala

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Profile

Aspiring AI Engineer with experience in developing machine learning models, NLP systems, and computer vision applications. Proven ability to build and deploy AI solutions in real-world environments.

Education

National School of Engineers of Sousse Sep 2023 – Present

- Engineering Degree in Applied Computer Science

Preparatory School of Engineering of Nabeul Sep 2021 – Jun 2023

- Preparatory Diploma in Technological Studies

Experience

AI Engineer Intern, Nehos Groupe, Sousse Jul 2025 – Sep 2025

- Designed and developed **ML and NLP models** for resume analysis, job-candidate matching, chat-bot assistance for employers and recruiters.
- **Tech Stack**: Python (FastAPI, scikit-learn, SpaCy, Google Generative AI), MySQL, React.js, Tailwind CSS, Express.js.

Software Engineer Intern, AKFER TUNISIA, Sousse Jun 2024 – Aug 2024

- Engineered a full-stack **collaborative web application** for employee management, productivity tracking, and real-time communication.
- **Tech Stack**: React.js, Express.js, Tailwind CSS, Socket.io, MySQL.

Projects

Thermal Image Classification using Few-Shot Learning github.com/fedikh/Few-Shot-Learning

- Built a **Few-Shot Learning model** using Prototypical Networks to classify defects in thermal images of induction motors.
- Implemented with **PyTorch** for efficient 5-shot classification, achieving high accuracy with limited data.

Cultural Algorithms for Image Processing github.com/fedikh/CulturalAlgo

- Implemented **cultural algorithms** integrated with machine learning for medical image segmentation.
- Designed belief spaces to guide evolutionary optimization, improving accuracy by **8%** over classical methods.

Plant Doctor – Plant Disease Detection github.com/fedikh/plant_doctor

- Developed a **machine learning model** to detect plant diseases from leaf images for a cross-platform mobile app.
- The model was integrated into a **Flutter** application for smooth UI and real-time camera detection.

Technologies

Languages: Python, Java, JavaScript, Objective-C, SQL

Frameworks & Libraries : PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, SpaCy, FastAPI, Seaborn, OpenCV, Flask, Django, Socket.IO, React.js, Node.js, Tailwind CSS

Databases : MySQL, PostgreSQL, MongoDB | **Tools** : Docker, MLflow, Git, GitHub, Postman

Certifications

- **Nvidia**: Fundamentals of Deep Learning | Accelerating Data Engineering Pipelines | Building LLM applications with prompt engineering
- **DeepLearning.AI**: Supervised Machine Learning | **IBM**: Introduction to Artificial Intelligence (AI)

Languages

Arabic: Native

English: Intermediate (working proficiency)

French: Intermediate (working proficiency)